Programming in the Era of AI

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Where I am coming from

- Author of high-performance libraries integrated into major browsers, runtimes and standard libraries: **simdutf**, **fast_float**, **simdjson**, **Roaring Bitmaps**.
- Among the top 2% of scientists worldwide (Stanford/Elsevier), 100+ peer-reviewed papers.
- Among the 1000 most-followed developers on GitHub.
- **And: I have not typed most of my code by hand in over a year.**

Preface

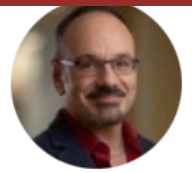
Back in the 1990s

The singularity

> *« I believe that the creation of greater than human intelligence will occur during the next thirty years. (I'll be surprised if this event occurs before 2005 or after 2030.) »*
>
> Vernor Vinge, 1993

Part 1

Where we actually are



Daniel Lemire
@lemire

Boost

What fraction of your code is typed by hand (no AI)?

none (vibe coding)	21.7%
20%	22.5%
50%	19.2%
all (old school)	**36.7%**

1,023 votes · Final results

7:19 PM · Mar 15, 2026 · **13.5K** Views

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Boost

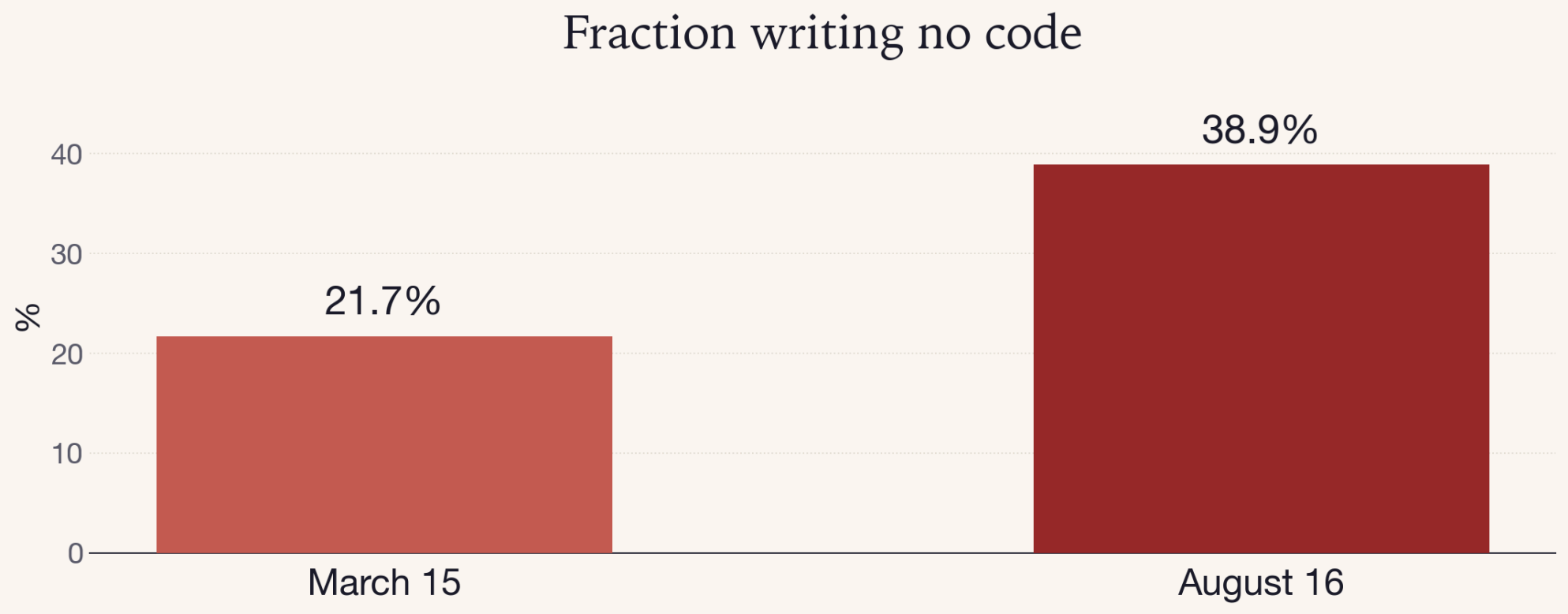
What fraction of your code is typed by hand (no AI)?

none (it is the future)	**38.9%**
20%	21.6%
50%	14.8%
all (old school)	24.6%

943 votes · Final results

2:58 PM · Aug 16, 2026 · **6,001** Views

What that poll really shows

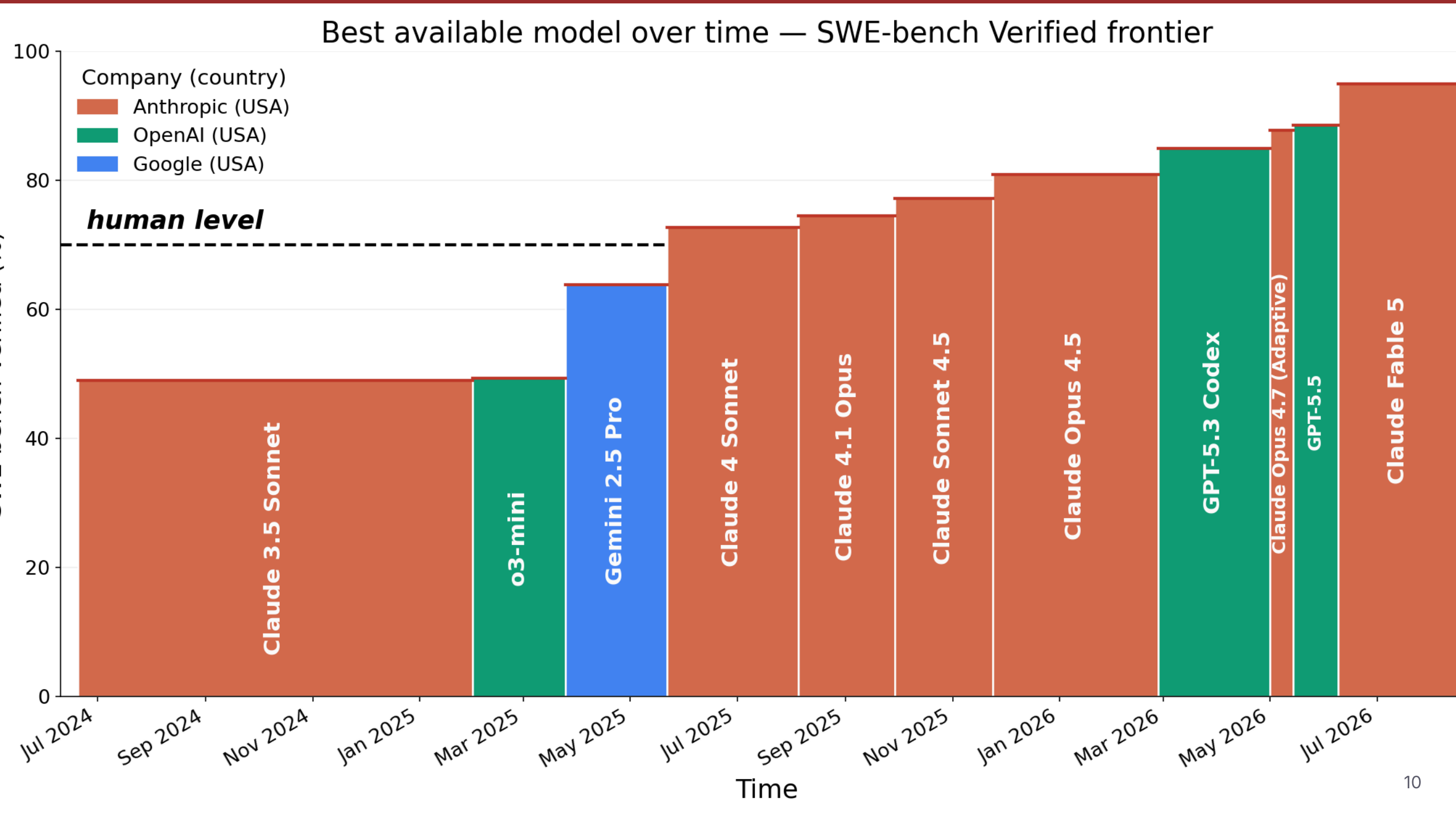


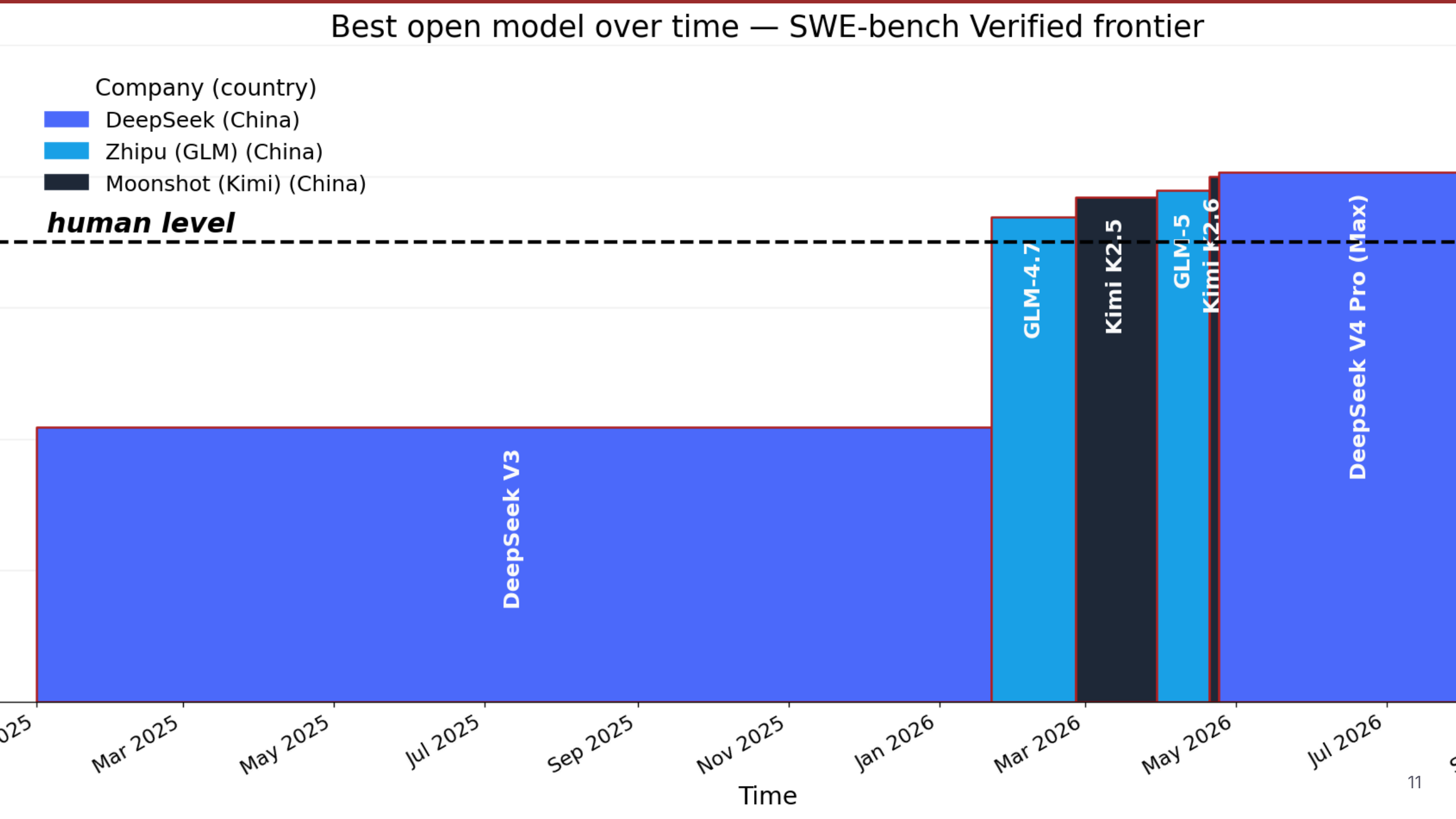
- this is a practice change, not a tool upgrade.

SWE-bench Verified

- Real GitHub issues from real open-source Python repositories.
- The model gets the repository, the issue text, and nothing else.
- Success = the project's own **hidden tests** pass afterwards.
- 500 tasks, human-validated as solvable.

It is not a quiz. It is a work order.


Best available model over time — SWE-bench Verified frontier
Company (country)
Anthropic (USA)
OpenAI (USA)
Google (USA)
human level
100
80
60
40
20
0
Claude 3.5 Sonnet
o3-mini
Gemini 2.5 Pro
Claude 4 Sonnet
Claude 4.1 Opus
Claude Sonnet 4.5
Claude Opus 4.5
GPT-5.3 Codex
Claude Opus 4.7 (Adaptive)
GPT-5.5
Claude Fable 5
Jul 2024
Sep 2024
Nov 2024
Jan 2025
Mar 2025
May 2025
Jul 2025
Sep 2025
Nov 2025
Jan 2026
Mar 2026
May 2026
Jul 2026
Time


Best open model over time — SWE-bench Verified frontier
Company (country)
DeepSeek (China)
Zhipu (GLM) (China)
Moonshot (Kimi) (China)
human level
DeepSeek V3
GLM-4.7
Kimi K2.5
GLM-5
Kimi K2.6
DeepSeek V4 Pro (Max)
Mar 2025
May 2025
Jul 2025
Sep 2025
Nov 2025
Jan 2026
Mar 2026
May 2026
Jul 2026
Time

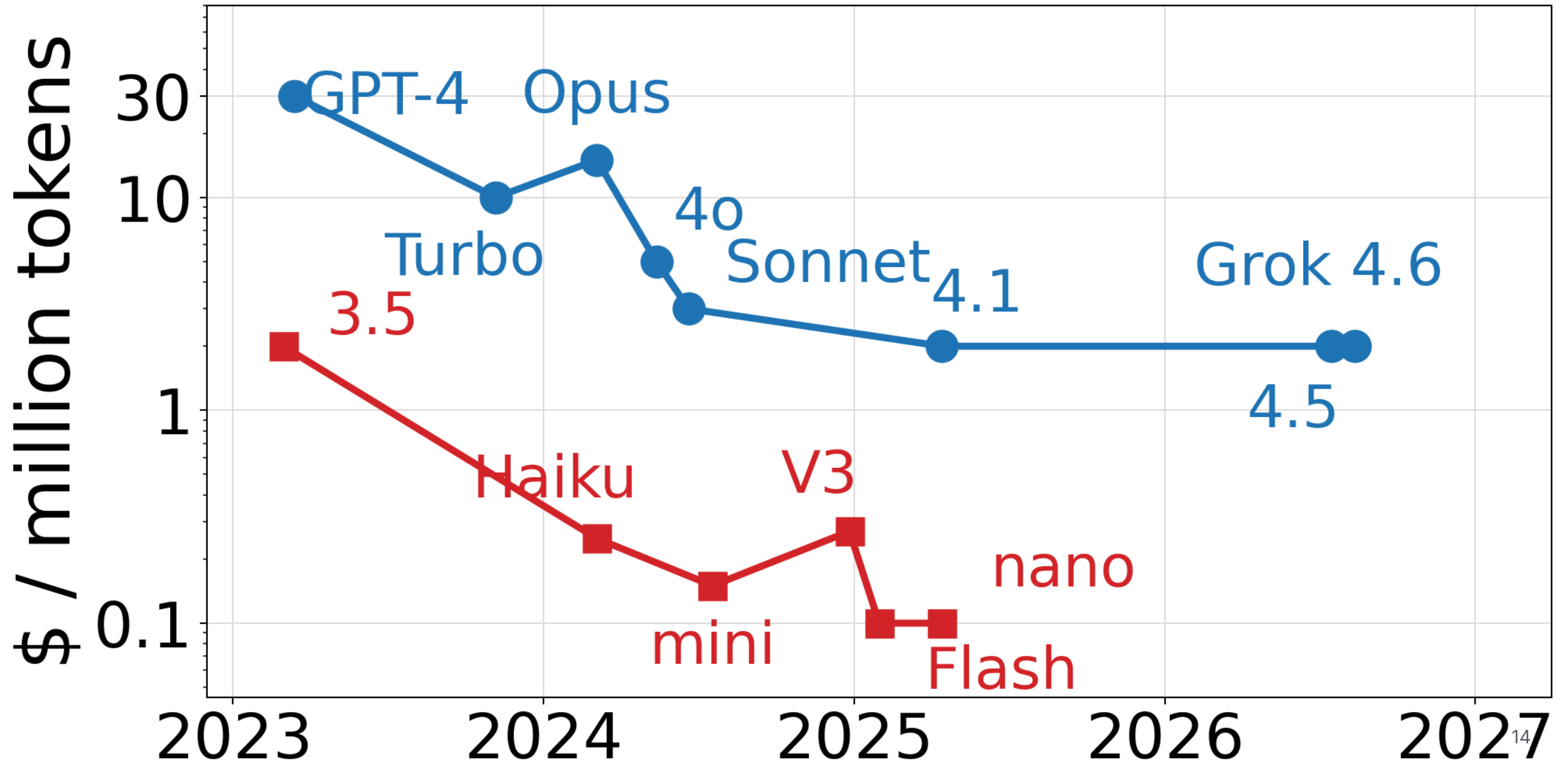
Superhuman

- Top systems now resolve **~90%** of SWE-bench Verified tasks.
- A skilled human given the same isolated issue, the same repository, and no colleagues, does **not** do better.
- The machine does it in minutes, in parallel, for a few dollars.

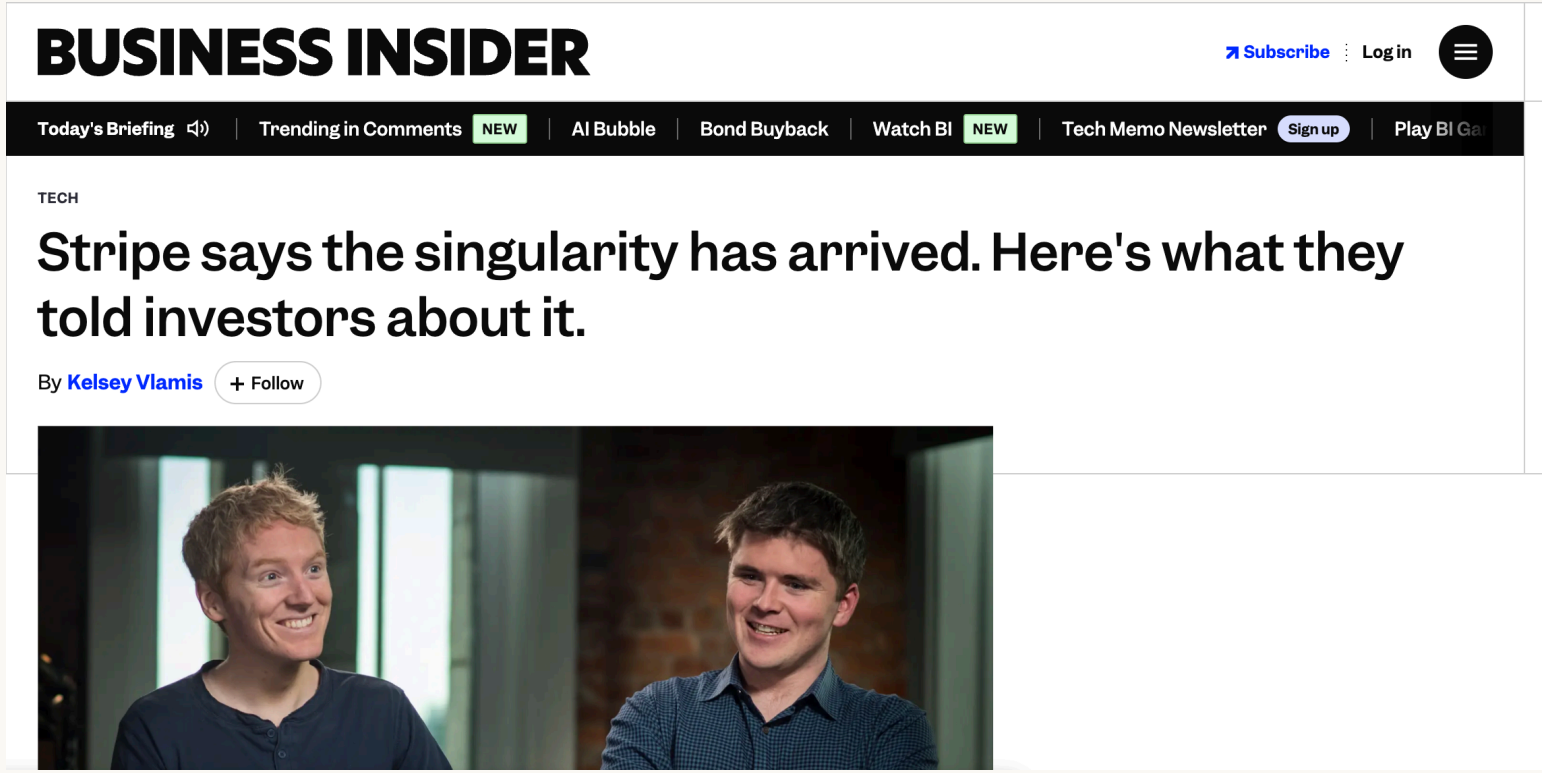
Open weights are close behind

- The leading open-weight models sit within a few points of the leading proprietary models.
- They run on hardware you can own, or rent by the hour.

Token price at launch



> *January 1st marked the beginning of the singularity.*
>
> —Stripe, August 19, 2026



People often talk about this concept called AGI, meaning artificial general intelligence, that is, an AI as intelligent as a person. I actually think we crossed that threshold about three months ago.

— Marc Andreessen, May 19, 2026

Three companies, founded within a decade

COMPANY	FOUNDED	VALUATION	DATE
OpenAI	2015	$852 billion	Mar. 2026
Anthropic	2021	$965 billion	May 2026
xAI	2023	$250 billion	Feb. 2026
Sum		**$2.07 trillion**	

COMPANY	VALUATION	DATE
Intel	$608 billion	May 2026
Oracle	$580–630 billion	May 2026
Manulife	$64 billion	May 2026
Bombardier	$21 billion	May 2026
Quebecor	$10–11 billion	May 2026

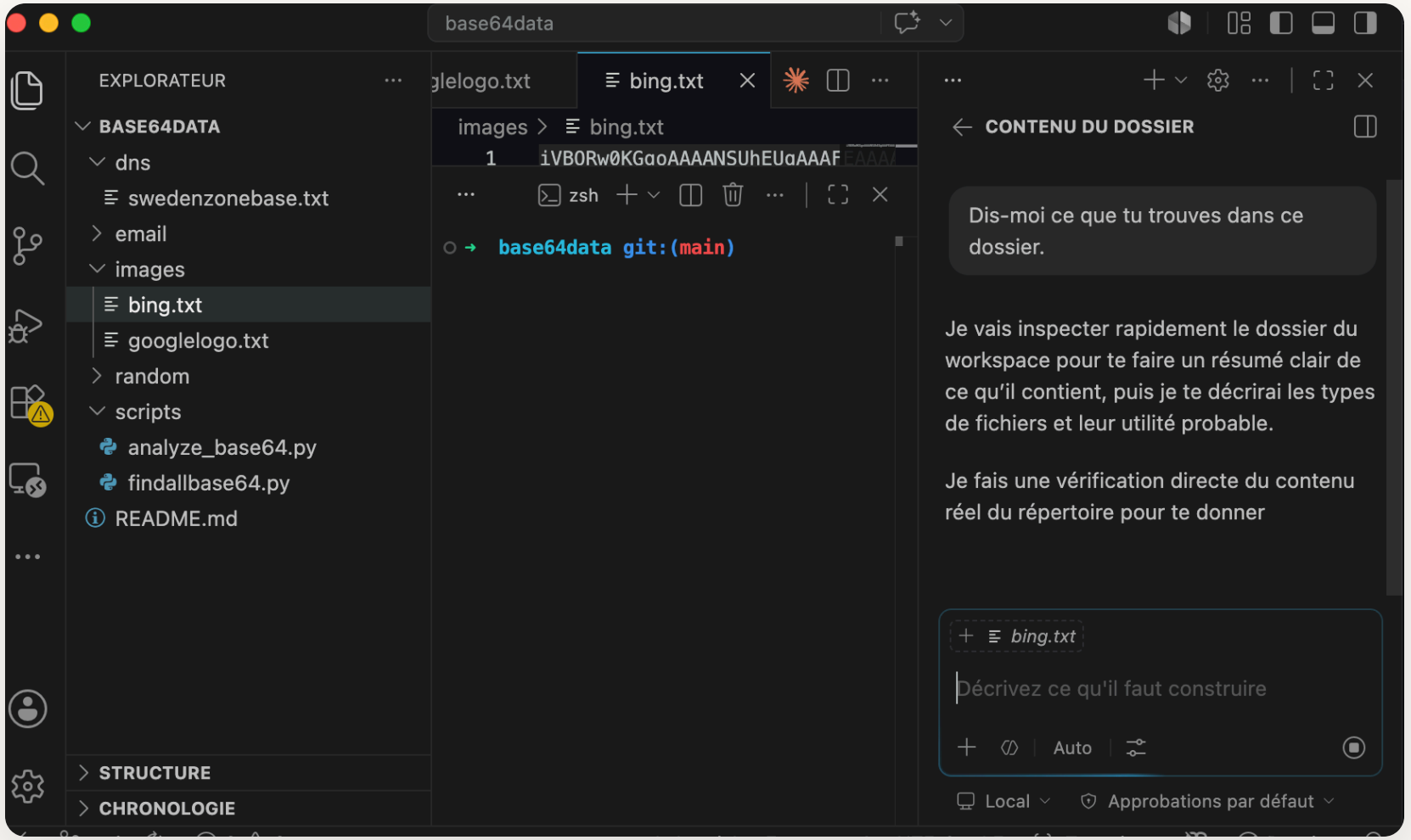
All Canadian public companies, excluding banks: **$2.6 trillion**


Run-rate revenue
Run-rate revenue ($bn)
$50bn
$40bn
$30bn
$20bn
$10bn
$0bn
$9bn
$14bn
$30bn
$47bn
Dec 31 2025
Feb 12 2026
Apr 1 2026
May 7 2026

Part 2

What makes an agent work

2024: autocomplete



GitHub Copilot + Visual Studio Code

- In-context code suggestion
- Whole-function generation
- Chat inside the IDE

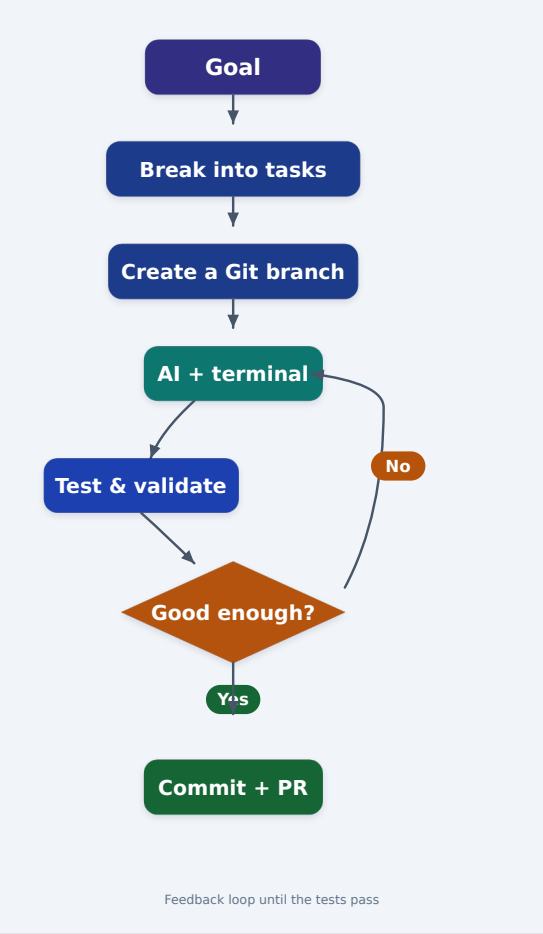
Useful. But **you** were still the loop.

Agentic AI

An agent can:

- Understand a goal
- Break it into steps
- Use tools (shell, web search, code execution, APIs)
- Observe the result and **adapt**

The model stopped being an oracle and became a worker.



Feedback loop until the tests pass

Why the command line won

- It is textual and deterministic.
- It exposes powerful, composable tools: `git`, `curl`, `jq`, `grep`, `find`, `awk`, compilers, test runners.
- Code can be executed directly, and the output read back.
- It is scriptable and observable.

The terminal is the richest tool API ever built, and it already exists.

Why Git matters more than ever

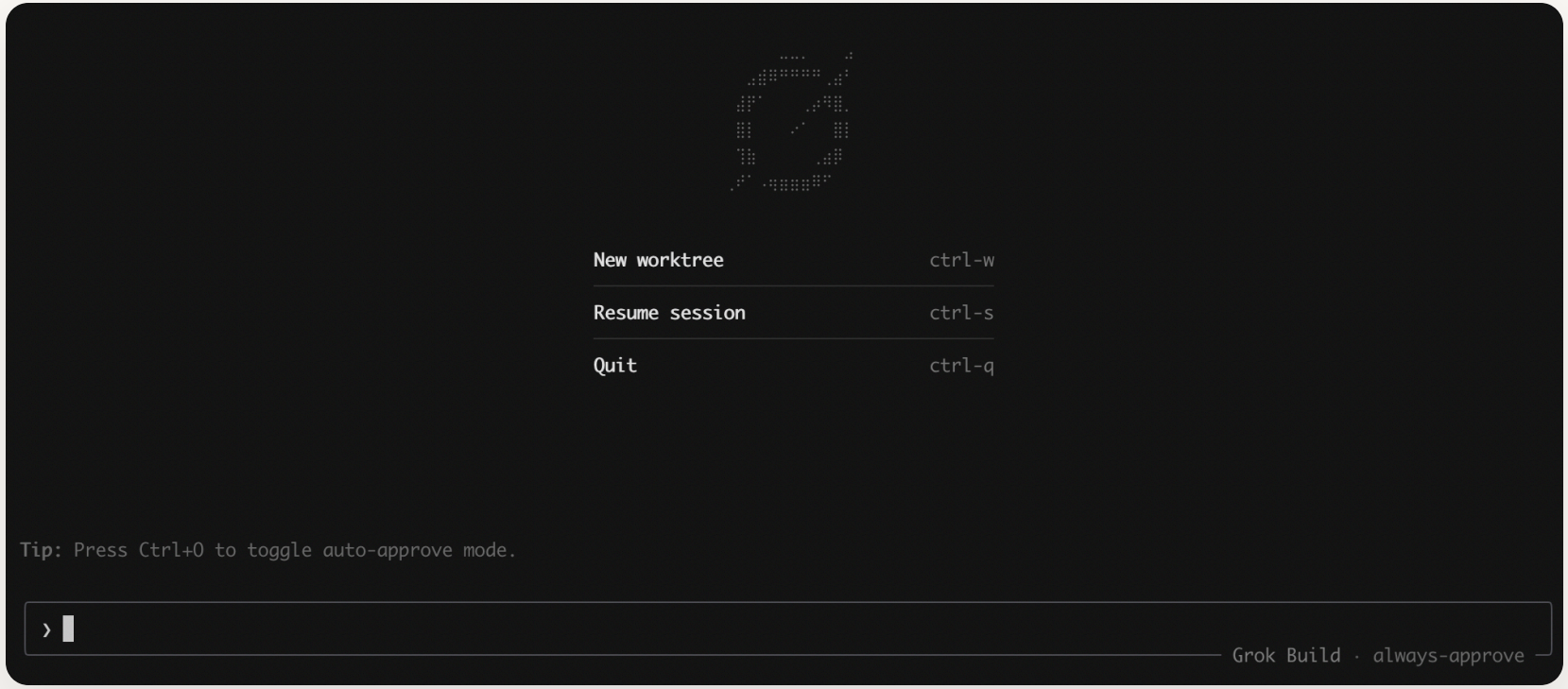
- **Cheap branching**: `git switch -c feature/agent-1`, and `git worktree` for parallel work.
- **Fearless iteration**: atomic commits plus `git reset`.
- **Review and validation**: `git diff`, `git log`, `git blame`.
- **An undo button** when the agent goes in a bad direction.

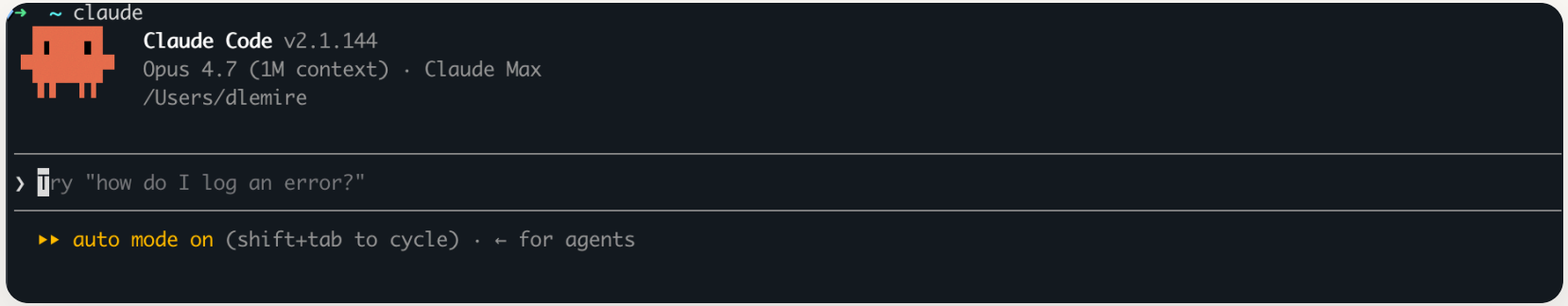
Git is the safety harness that makes delegation rational.

Why Markdown

```
## Introduction to Java (example)

We can define a variable in Java like this.

Notice the different elements of the syntax.

- Type
- Variable name
- Assigning a value
```

Structured enough for a machine, readable enough for a human, diffable in Git.


New worktree ctrl-w
Resume session ctrl-s
Quit ctrl-q
Tip: Press Ctrl+O to toggle auto-approve mode.
Grok Build · always-approve


~ claude
Claude Code v2.1.144
Opus 4.7 (1M context) · Claude Max
/Users/dlemire
❯ Try "how do I log an error?"
▸▸ auto mode on (shift+tab to cycle) · ← for agents

Constraints

Benchmarks and tests as the steering wheel

The central problem

A language model will produce something plausible **every single time**.

It has no way, on its own, to know whether it is right.

So the entire engineering problem becomes: **give it something it can check.**

The hierarchy of constraints

From weakest to strongest:

1. "Looks good to me"
2. A type checker or linter
3. A unit test
4. A **property** test or differential test against a reference
5. A **fuzzer**
6. A **benchmark** with a numeric target

Each level lets you delegate more and supervise less.

Tests are no longer just for regressions

- Historically: tests protect code you already wrote.
- Now: tests **specify** code that does not exist yet.
- The test suite is the contract that the agent iterates against.

Write the test first — not for purity, but because it is the only instruction the machine cannot talk its way out of.

A benchmark is an even better constraint

A test says *correct / incorrect*.

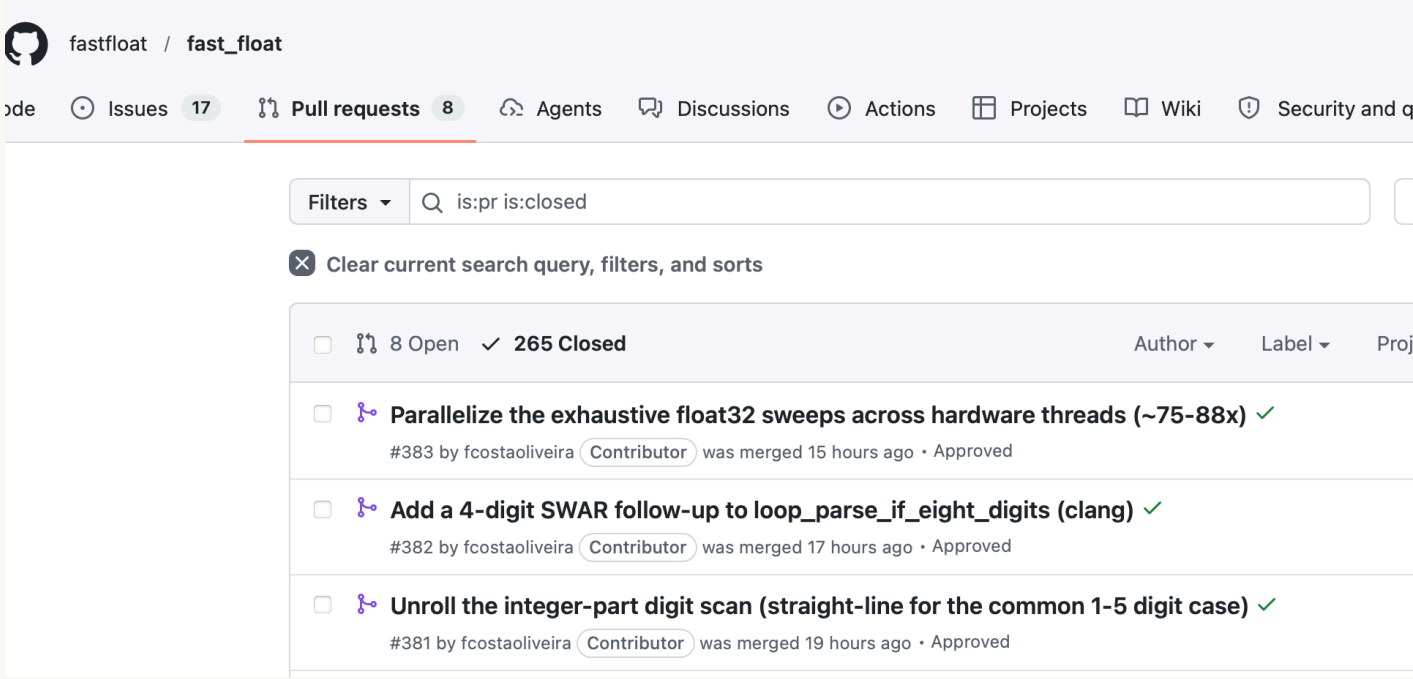
A benchmark says *how much better*.

```
Make `parse_number` faster. Run `make bench` after every
attempt. Do not stop until cycles/byte drops below 0.20
and the differential fuzzer still passes.
```

The agent now has a gradient to climb, and a stopping condition.

It works on hard code

- `fast_float` is used by Chrome, Safari, GCC, Rust, Go, MySQL.
- It has been tuned by hand for five years.
- An agent, given the benchmark, found **two independent 10% improvements**.



Why that surprised me

- I did not believe there was 20% left.
- The agent was not smarter than the contributors. It was **more patient**.
- It tried hundreds of variants overnight and kept the ones the benchmark liked.

Search plus a good objective function beats intuition.

Differential testing: the workhorse

```
Here is a slow, obviously-correct reference implementation.
Here is a fuzzer that feeds random inputs to both and
compares outputs, including edge cases: empty input,
one byte, unaligned buffers, invalid encodings.

Write a fast version. Run the fuzzer after every change.
```

> You are not reviewing the code. You are reviewing **the contract**.

> *If I cannot state how I would check the result, I do not delegate it.*

Context

The scarce resource

Context windows are large now

A million tokens is roughly a 3000-page book. So the problem is solved?

No. A large context is a large *haystack*.

- Attention dilutes: the more irrelevant material you include, the more the model misses the relevant part.
- Stale information competes with fresh information — and looks identical.
- Cost and latency scale with what you keep.
- The failure is silent: you get a confident answer built on the wrong file.

Context management: the moves

1. **Curate, don't dump.** Retrieve the three right files, not the repository.
2. **Persist the durable facts** in a file the agent always reads.
3. **Externalize state** to disk and to Git, not to conversation history.
4. **Isolate**: give a sub-task its own fresh context.
5. **Compact deliberately**: summarize and restart rather than letting a session rot.

A project memory file

```
# CLAUDE.md

## Build
`cmake -B build && cmake --build build -j`

## Test
`ctest --test-dir build --output-on-failure`

## Rules
- C++17. No exceptions in the hot path.
- Every SIMD kernel needs a scalar reference + a fuzzer.
- Never edit files under `third_party/`.
```

Written once. Read at the start of every session. This is the highest-leverage file in the repository.

Permissions

```
{
  "permissions": {
    "allow": ["Read", "Write", "Edit", "Bash(git status)", "Bash(git commit -m:*)"],
    "deny": ["Read(.env*)", "Bash(rm -rf /)", "Bash(sudo:*)"],
    "ask": ["Bash(git push --force:*)", "Bash(docker run:*)"]
  }
}
```

`~/.claude/settings.json`

allow, deny, ask

- **allow**: automatically authorized (low risk, frequent)
- **deny**: forbidden at all times (secrets, destructive operations)
- **ask**: sensitive actions requiring explicit confirmation

The goal is not to be safe. The goal is to be safe enough that you stop supervising every keystroke.

MCP (Model Context Protocol)

- Connects a model to external tools
- Standardizes tool access across vendors
- Examples: Git, Slack, Oracle, PostgreSQL, SSH, Google Drive, your internal API

If the command line is the universal tool API, MCP is the one for everything that is not a command line.

Extending capabilities securely

- Expose **only** the strictly necessary actions in the MCP server
- Confine every path to a sandbox directory
- Traceability: log MCP calls and audit them regularly

For an SSH/SFTP skill: read and write in a single folder, and every destructive operation goes in `ask` mode.

Example MCP server: `server.py`

- Starts an MCP server named `ssh-files`
- Reads `credentials.json` (host, username, remote directory)
- Exposes `upload_file`, `download_file`, `list_files`, `make_dir`, `delete_file`, `delete_dir`
- All paths confined to the remote directory (no sandbox escape)
- Verifies SSH host keys

```
claude mcp add ssh-files server.py --scope user
```

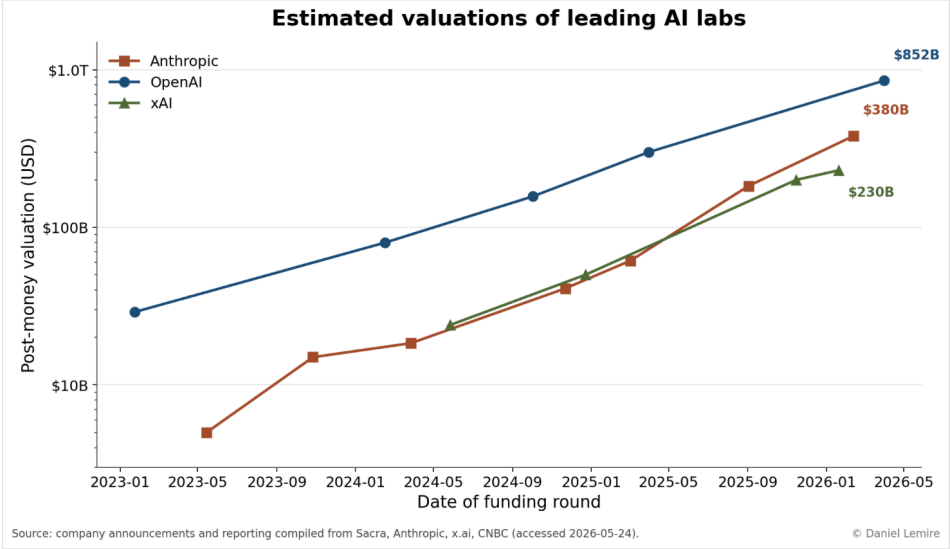
```
claude mcp list

  - claude.ai Google Drive — authentication required
  - ssh-files (./ssh-mcp/server.py) — connected
```

```
/plotdata Estimated market value of Anthropic, OpenAI, and xAI.

Then upload the result using the ssh-files MCP into the
corresponding directory, create a nice index.html file,
and give me the URL.
```

https://lemire.me/plot_data/ai-lab-valuations/

Estimated valuations of leading AI labs

Post-money valuations of Anthropic, OpenAI and xAI from publicly reported funding rounds.



Log-scale post-money valuation by funding round. Latest data points: OpenAI $852B (Mar 2026), Anthropic $380B (Feb 2026), xAI $230B (Jan 2026).

Sources

- Anthropic — Series G at $380B post-money (Feb 12, 2026)
- Anthropic — Series F at $183B post-money (Sep 2025)
- Sacra — Anthropic revenue, valuation & funding
- CNBC — OpenAI closes funding round at $852B valuation
- Sacra — OpenAI revenue, valuation & funding
- xAI — Series E ($20B)
- Sacra — xAI revenue, valuation & funding

Orchestration

Many agents at once

Why parallelism, concretely

- A single agent run is minutes of wall-clock time, and most of it is spent waiting on a build or a test suite.
- Running four agents costs four times the tokens and roughly zero extra minutes.

`git worktree` is the enabling primitive:

```
git worktree add ../work-a -b feature-a
git worktree add ../work-b -b feature-b
```

Each agent gets its own directory and its own branch. Same repository, no shared working tree, no file collisions. Merge or discard independently.

Pattern 1: fan-out over independent work

One task list, one agent per item.

- Migrate 30 files to a new API
- Add tests to 12 modules
- Port a kernel to five instruction sets

Requirement: the items must not touch the same files.

Pattern 2: the panel

Ask **N** agents the same question independently, then compare.

- Three designs for the same feature, then pick one.
- Three reviewers on the same patch, keep findings that two of them agree on.

Independent samples catch what one confident sample does not.

Pattern 3: adversarial verification

The generator and the critic must not be the same context.

```
Agent A: implement the feature until the tests pass.
Agent B: given only the diff, try to prove it is wrong.
         Find an input where it misbehaves.
```

A model asked to defend its own work will defend it.

What does *not* parallelize

- Anything with a shared, mutable, non-mergeable resource: one database, one port, one flaky integration test.
- Work where step 2 genuinely needs the answer to step 1.
- Anything where merging costs more than doing.

And you still have to read the results. That does not parallelize at all.

The honest accounting

	COST
Tokens	multiplied by the number of agents
Wall-clock	roughly unchanged
Your review time	**multiplied by the number of agents**

Parallelism buys throughput only if verification is automated.
Otherwise you have built a machine for generating homework.

Part 3

Field reports

Part 4

What we must change

Teaching: what stops working

- Take-home programming assignments as an assessment of *coding*.
- "Implement a linked list" as a filter.
- Grading the artifact instead of the reasoning.
- Any exercise whose specification is complete enough to paste into a chat box.

If a task is fully specified, it is already automated.

Teaching: what becomes essential

- **Reading** code critically — the skill formerly gained by writing it.
- **Specification**: turning a vague need into a checkable statement.
- **Testing**, fuzzing, property-based thinking.
- **Measurement**: benchmarks, statistics, performance counters.
- **Systems fluency**: the shell, Git, build systems, deployment.
- Knowing what is **hard**, so you notice when the answer is too easy.

Assessment has to move

- Oral defence of a design decision.
- Review a patch, in the room, and justify the review.
- Give students an agent and a hard problem, and grade the **process**.
- Assume the tool is present, then ask for something it cannot do alone.

We stopped grading arithmetic when calculators arrived. We did not stop teaching mathematics.

Work organization is the real bottleneck

Two roles are emerging:

- **The coder-analyst** — closest to the problem, now able to build the tool themselves.
- **The architect-programmer** — sets constraints, defines interfaces, owns the tests, reviews.

The team structure built around "one specification, thrown over a wall, then six months of implementation" no longer matches the cost structure.

Review becomes the constraint

- Generating a 2000-line patch costs minutes.
- Reviewing a 2000-line patch costs a day.
- The queue moves to the humans.

Everything that makes review cheaper is now the highest-value engineering investment: small diffs, strong tests, clear invariants, automated checks.

Research: what to prioritize

- **Verification at scale**: how do we gain confidence in code no human read line by line?
- **Better objective functions**: benchmarks that capture what we actually want.
- **Context and retrieval**: what to show a model, and when.
- **Orchestration**: when does a second agent help, and when is it noise?
- **Empirical software engineering**: we have a new, measurable subject and very little data.

The bad news

- Models are confident when they are wrong, and the failure is invisible in the diff.
- Generated code raises volume: more code, more surface, more supply-chain exposure.
- Prompt injection is a real attack: an agent that reads the web and can run commands is a new class of target.
- Skills are eroding in the people who never had to acquire them the hard way.
- The economics of junior positions are genuinely under pressure.

None of this is a reason to opt out. All of it is a reason to build the checkers.

If you take five things home

1. The capability is real, it is cheap, and it is available to your competitors.
2. **Whatever you cannot check, you cannot delegate.** Build the checkers first.
3. Tests and benchmarks are no longer overhead — they are the steering wheel.
4. Context is the scarce resource: curate, persist, isolate, compact.
5. Parallel agents buy throughput only when verification is automated.

Questions?

Daniel Lemire — lemire.me (blog)

https://simdjson.org · https://roaringbitmap.org

https://simdutf.github.io/simdutf/ · https://fastfloat.github.io/fast_float/